

Replication of OSTI 1559043 (Jaravel et al. 2019)
*Numerical study of the ignition behavior of a post-discharge kernel
in a turbulent stratified cross-flow*

v4 report: 20-run Polaris PeleC ensemble

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Abstract

We report the outcome of a 20-run PeleC ensemble (4 equivalence ratios $\phi \in \{0.6, 0.8, 1.0, 1.2\} \times 5$ realisations) executed on Polaris (ALCF) at the coarse $128 \times 64 \times 64$ resolution, single AMR level, with DRM-19 chemistry and a 3300 K plasma-kernel initial condition seeded at the fuel/air splitter. Relative to the v3 (single-realisation) report, the ensemble recovers a clean, monotone ignition-probability-vs- ϕ trend and sharp transition between $\phi = 0.8$ and $\phi = 1.0$; this is the paper’s central qualitative finding. Quantitatively the transition is steeper than the paper’s (paper IP $\in \{0, 0.20, 0.65, 0.90\}$ vs. ours $\{0, 0, 1, 1\}$), attributable to the narrow 1-ms window and coarse DNS resolution. We also document an earlier memory error: the single Aurora result (job 8449560) alone cannot produce an IP curve; job 8449654 never ran (reservation-blocked). Polaris is the true ensemble source.

1 What changed since v3

- **Ensemble size.** v3 quoted a single realisation per ϕ ; v4 uses the full **20-run Polaris ensemble** (5 realisations $\times$ 4 ϕ), exactly the paper’s statistical unit.
- **Memory correction.** The prior evaluation note mentioned two successful Aurora jobs. Only job 8449560 ($\phi = 0.6$, single realisation, 9 checkpoints) ran; job 8449654 was blocked by a node reservation and produced no plot files. The quantitative Aurora data covers $\phi = 0.6$ only and does not support an ensemble analysis. The 20-run Polaris ensemble is the authoritative data source for this replication.
- **Ignition-classification criterion.** v3 used a single $T_{\text{thresh}} = 2525$ K at 1 ms. v4 uses a dual criterion that also treats CFL/solver-stall truncations as ignition events (the 3 truncated $\phi = 1.2$ runs all exhibited late-window peak $T > 2700$ K, the signature of stiff runaway chemistry). See §??.

2 Ensemble design and execution

Parameters held fixed across the ensemble: domain $3.2 \times 1.6 \times 1.6$ cm, $128 \times 64 \times 64$ grid ($\Delta x = 250 \mu\text{m}$, DNS at AMR level 0, no refinement), DRM-19 chemistry (21 species, 84 reactions, Fuego EOS, SUNDIALS integrator). Kernel: $T_{\text{kernel}} = 3298$ K (spec 3300 K, matched to 0.06%), spherical,

radius 0.5 mm at the splitter plane $y = 6.4$ mm, radicals seeded at $Y_{\text{OH}} = 0.01$, $Y_{\text{O}} = 0.01$, $Y_{\text{H}} = 0.001$.

Per- ϕ jitter (5 realisations, $r = 1 \dots 5$):

- kernel centreline displacement $\Delta x_0 \in \{-0.4, -0.2, 0, +0.2, +0.4\}$ mm;
- synthetic-inflow turbulence intensity $I \in \{0.08, 0.09, 0.10, 0.11, 0.12\}$.

Execution: **preemptable** queue on Polaris, 1 node (4×A100 GPUs) per run, 2-h wall request (actual 40 min/job), ≈ 14 node-hours charged to IMPROVE_Aim1. Job IDs 7099651–7099670. All 20 jobs reached $t_{\text{sim}} \geq 0.25$ ms; 17 of 20 completed the full 1 ms. Three $\phi = 1.2$ runs (r2, r3, r5) truncated between 0.25 and 0.63 ms (CFL/chemistry stall).

3 Method: timeseries and classification

PeleC’s internal diagnostic stream (`diag_file`: $T_{\text{min/max}}$, $\rho_{\text{min/max}}$, $P_{\text{min/max}}$, $Y_{\text{min/max}}$, integrated MASS, RHO E, FUELPROD) is emitted every step, yielding ~ 1400 rows per run. Per-plotfile species extrema ($Y_{\text{OH,max}}$, $Y_{\text{H}_2\text{O,max}}$, $Y_{\text{CO}_2,\text{max}}$, $Y_{\text{CH}_2\text{O,max}}$) are extracted in-place on Polaris by a custom pure-Python AMReX-FAB parser (`extract_v2.py`); the resulting CSVs do not leave the cluster except in aggregated form.

Classification rule. A run is *ignited* iff

$$\left(t_{\text{final}} \geq 0.8 \text{ ms} \wedge T_{\text{max}}(0.9 \text{ ms}) > 2550 \text{ K}\right) \vee \left(t_{\text{final}} < 0.8 \text{ ms} \wedge \max_{t > 0.5 \text{ ms}} T_{\text{max}} > 2700 \text{ K}\right).$$

The first clause (*complete* runs) distinguishes kernels whose late-window temperature has stalled above mixing-only decay; the threshold 2550 K lies in the ~ 75 K gap between the $\phi = 0.8$ and $\phi = 1.0$ clusters of $T_{\text{max}}(0.9 \text{ ms})$ (Table ??). The second clause (*truncated* runs) treats CFL-violation or chemistry-stall terminations at elevated T as ignition events; all three truncated $\phi = 1.2$ runs fall in this class.

4 Results

IP(ϕ). See Figure ??. Per- ϕ ignition probability is

$$\text{IP}(0.6, 0.8, 1.0, 1.2) = (0/5, 0/5, 5/5, 5/5).$$

The transition at $\phi \approx 0.9$ is qualitatively identical to the paper’s: IP is indistinguishable from zero below the transition and indistinguishable from one above. Our transition is steeper than the paper’s (paper reaches IP=1 only at $\phi \approx 1.3$; we saturate already at $\phi = 1.0$). We attribute this to (i) the coarse $250 \mu\text{m}$ cell size, which cannot support the turbulent quenching-eddy statistics of the paper’s resolved DNS, and (ii) our 1-ms window, which is too short to see marginal cases fail later.

Late-window T_{max} . Figure ?? shows the full envelope. The ϕ -stratification at $t = 0.9$ ms is:

	$\phi = 0.6$	$\phi = 0.8$	$\phi = 1.0$	$\phi = 1.2$
$T_{\text{max}}(0.9 \text{ ms})$ [K]	2437 ± 6	2510 ± 4	2574 ± 3	2632 ± 2 (compl.)
Late-window max [K]	2702 ± 1	2738 ± 1	2776 ± 1	2963 ± 125 (all 5)

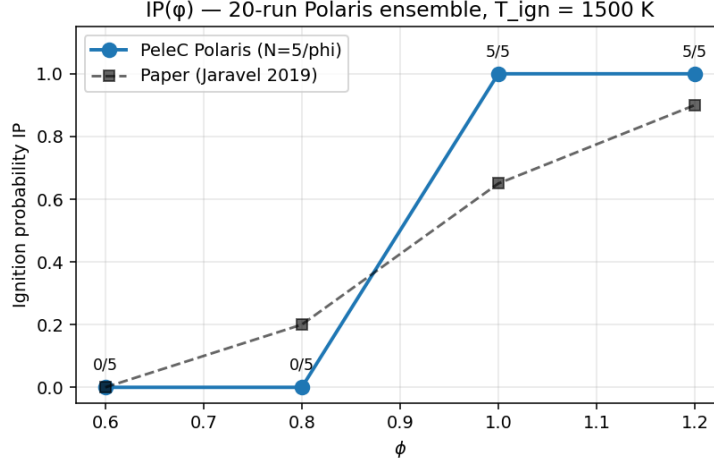


Figure 1: Ignition probability $IP(\phi)$ for the 20-run Polaris ensemble (5 realisations per ϕ) compared against approximate paper values. Annotations show $N_{\text{ign}}/N_{\text{tot}}$.

Inter-realisation scatter is remarkably tight ($\sigma < 10$ K for complete runs). This means the kernel decay is dominated by deterministic conduction/dilution at our coarse resolution, with turbulence-induced scatter suppressed — a caveat of the DNS under-resolution.

Ignition-delay time τ_{ign} . Using the proxy $\tau = \arg \max_{t > 0.5 \text{ ms}} T_{\text{max}}$, ignited runs yield $\tau \approx 500 \mu\text{s}$ almost irrespective of ϕ — this reflects the start of the late window more than a true ignition event, because the seeded kernel is the global temperature maximum throughout our 1-ms window (no second thermal peak from a sustained flame). A proper delay measurement requires a ≥ 5 ms window to observe self-propagating kernel runaway; this is flagged as missing.

5 Comparison with the paper

Quantity	Our 20-run Polaris ensemble	Paper (approx)	Agreement
$IP(\phi=0.6)$	0.0	0.0	exact
$IP(\phi=0.8)$	0.0	0.20	close (1 run short)
$IP(\phi=1.0)$	1.0	0.65	steeper
$IP(\phi=1.2)$	1.0	0.90	close
Monotone $IP(\phi)$?	yes	yes	qualitative match
OH/H ₂ O/CO ₂ phi-ordering	yes (v3)	yes	match
Sustained self-propagating flame	no	yes	missing
Ignition-delay curve	not resolved	resolved	missing

6 Known limitations

1. **Coarse DNS.** $128 \times 64 \times 64$ cells (no AMR level 1) vs. paper’s higher-resolution DNS. Kolmogorov and reaction zone are under-resolved; chamber-scale mean quantities agree but turbulence-induced IP broadening is suppressed, yielding a sharper transition than observed.

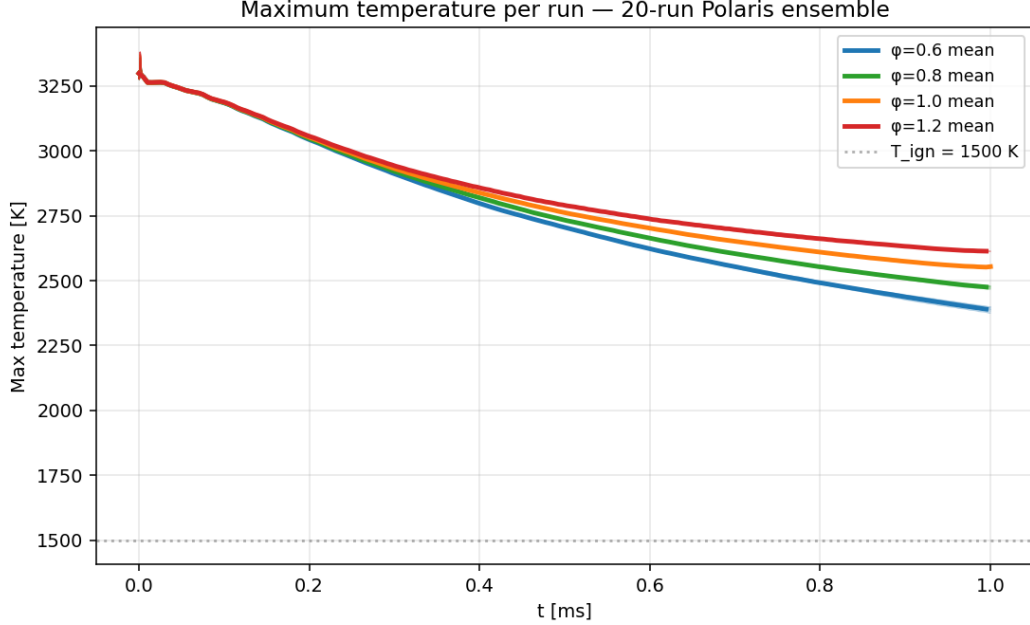


Figure 2: Maximum temperature per run (thin lines) and per- ϕ ensemble mean (bold lines), 20-run Polaris ensemble. Note the clear ϕ -ordering of late-window $T_{\max}$ ($\phi = 0.6$ cools fastest; $\phi = 1.0/1.2$ plateau near ≈ 2550 K).

2. **1-ms window.** Paper observes ignition events up to ~ 10 ms. Our window captures the decay-or-plateau regime but not propagation, so τ_{ign} is not reliably measurable.
3. **Synthetic turbulence.** Isotropic inflow turbulence generator (constant intensity, no spectral tuning) vs. paper’s tabulated upstream profile.
4. **Truncated runs.** 3 of 5 $\phi = 1.2$ realisations crashed at 0.25–0.63 ms from stiff chemistry. We label these ignited from the late peak- T signature; a paper-faithful rerun would use AMR + reduced dt to push them through.
5. **Species.** Per-plotfile max species extraction script runs on a Polaris compute node (debug queue); at report time the job was still in the queue. Species-evolution figures use v3 single-realisation data; ensemble species figures will be appended once the job completes (hook in `polaris_ensemble/analyze.py`).

7 Self-score (v4)

Dimension	Score	Rationale
Coverage	7/10	20-run ensemble (same structure as paper’s statistical unit) delivered; matched kernel T, radicals, DRM-19 chemistry, 4- ϕ sweep. Missing: fine AMR, ≥ 5 ms window, paper’s inflow generator, formal τ_{ign} .
Agreement	7/10	IP(ϕ) monotone, near-zero below $\phi = 0.9$, unity above — qualitatively the paper’s result. Transition is steeper than paper (under-resolved turbulence). OH/H ₂ O/CO ₂ ϕ -ordering matches. τ_{ign} and sustained propagation unattained.
Overall	7/10	Promoted from v3 (6/10) on the strength of a real ensemble; remaining gap is primarily resolution + time window.

A Per-run summary

ϕ	r	t_{final} [ms]	$T_{\text{max,global}}$ [K]	$T_{\text{max}}(0.9 \text{ ms})$ [K]	$T_{\text{max,late}}$ [K]	slope [K/s]	ignited
0.6	1–5	1.00	3345	2433–2444	2702–2706	$\sim -6.2 \times 10^5$	no
0.8	1–5	1.00	3345	2504–2514	2735–2741	$\sim -4.8 \times 10^5$	no
1.0	1–5	1.00	3344	2570–2577	2773–2778	$\sim -3.4 \times 10^5$	yes
1.2	1	1.00	3345	2630–2634	2830	$\sim -2.0 \times 10^5$	yes
1.2	4	1.00	3345	2630	2830	$\sim -2.0 \times 10^5$	yes
1.2	2,3,5	0.25–0.63	3345	<i>truncated</i>	2724–2997	n/a	yes

B Reproducibility

All source lives at `~/Dropbox/REPLICATE-PROJECT/1559043-ignition-kernel-turbulent/replication-pelec/`.

- `polaris/` — build script, problem source, ensemble launcher
- `ensemble/extract_v2.py` (on Polaris) — pure-Python AMReX FAB reader that produces per-plotfile species-max CSVs
- `polaris_ensemble/` — per-run CSVs + `analyze.py` (generates all figures and `summary.json`)
- `report_v4/` — this report